

# Computer Vision - Exercise 5

Jun.-Prof. Dr. Kira Maag, Ben Hamscher M.Sc., Arnold Brosch M.Sc.

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This exercise will be discussed the week of 8th of June.

## 1 Introduction

The following exercises cover perspective projection, epipolar geometry, and the essential and fundamental matrices.

## 2 Exercises

### Exercise 1: Triangulation and 3D Reconstruction

To reconstruct the 3D position of an observed point, we use two cameras with known projection matrices. Let the projection matrix of the first camera be

$$P_1 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

and the projection matrix of the second camera be

$$P_2 = \begin{pmatrix} 1 & 0 & 0 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \end{pmatrix}.$$

A 3D point  $X_{true}$  is observed in the first camera at the homogeneous image coordinates  $x_1 = (1, 2, 1)^\top$  and in the second camera at  $x_2 = (0, 2, 1)^\top$ .

1. Let the 3D point in homogeneous coordinates be  $X = (X, Y, Z, 1)^\top$ . Why can we not simply compute the 3D point using the pseudoinverse of the projection matrix, e.g.,  $X = P_1^+ x_1$ ?
2. To eliminate the unknown scale factor  $\lambda$  in the equation  $x_1 \simeq P_1 X$ , we can use the cross product constraint  $x_1 \times (P_1 X) = 0$ . Apply this constraint to the first camera and write down the first two independent linear equations.
3. Apply the same cross product constraint  $x_2 \times (P_2 X) = 0$  to the second camera and write down the first two independent linear equations.

4. Solve the resulting system of linear equations to find the Cartesian 3D coordinates  $(X, Y, Z)$  of the point.

## Exercise 2: Epipolar Constraint and Essential Matrix

Let two calibrated cameras observe a 3D point. The corresponding normalized image points are  $x$  and  $x'$ .

The essential matrix is defined as

$$E = [t]_{\times} R,$$

where  $R$  is a rotation matrix and  $t$  a translation vector.

1. Explain the geometric meaning of the epipolar constraint

$$x^{\top} E x' = 0.$$

2. What is the role of the skew-symmetric matrix  $[t]_{\times}$ ?
3. Show why  $E e' = 0$  holds and explain the meaning of  $e'$ .
4. Why does the essential matrix have rank 2?

## Exercise 3: Fundamental Matrix and Pixel Coordinates

Let  $p$  and  $p'$  be corresponding points in pixel coordinates. The fundamental matrix is defined as

$$F = K^{-\top} E K'^{-1}.$$

1. Derive the epipolar constraint in pixel coordinates:

$$p^{\top} F p' = 0.$$

2. Explain the role of the intrinsic calibration matrices  $K$  and  $K'$ .
3. What is the geometric interpretation of the epipolar line

$$l' = F^{\top} p?$$

4. How does the fundamental matrix differ from the essential matrix in terms of required knowledge about the cameras?